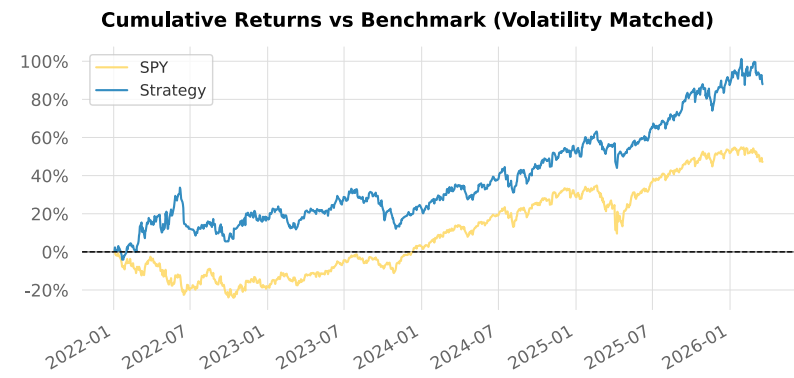
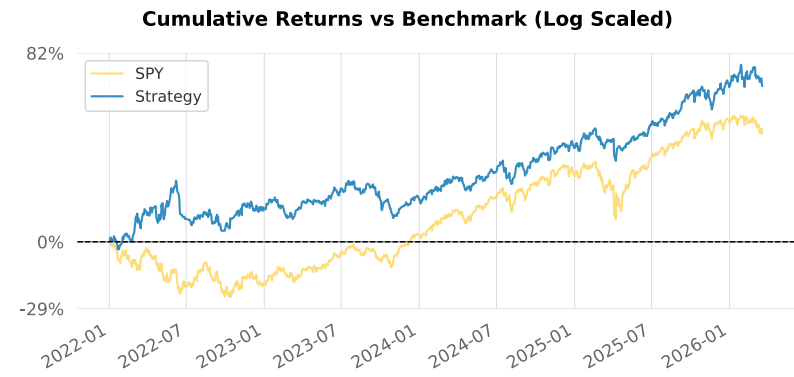
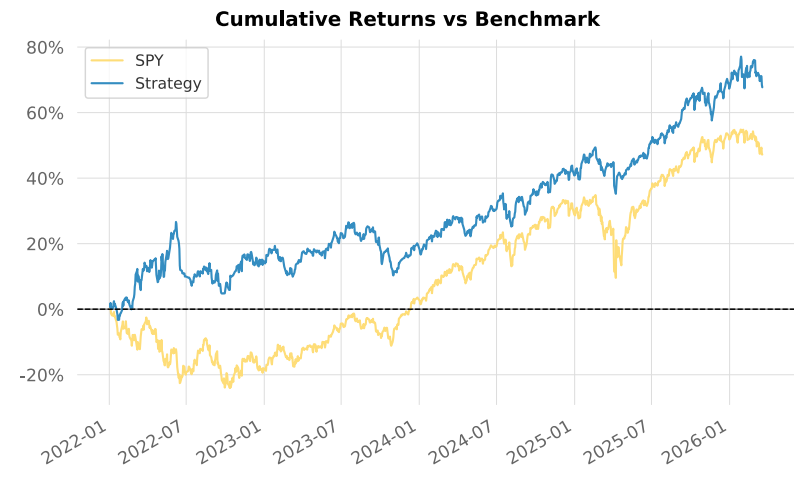


Multi-Domain Swarm V8 - Out-Of-Sample (2022-2026)

(Compounded) 3 Jan, 2022 - 19 Mar, 2026 (matched dates)

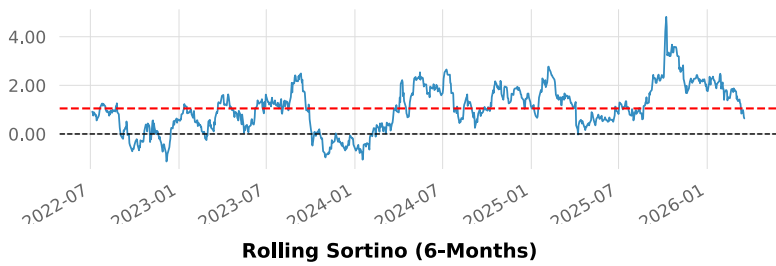
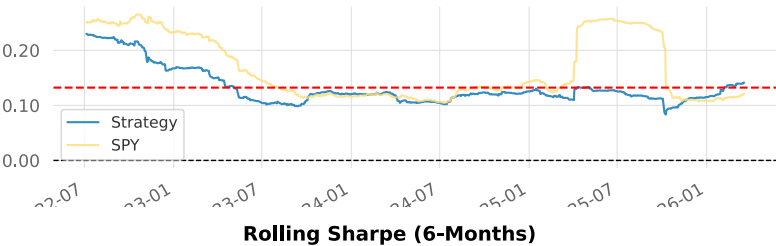
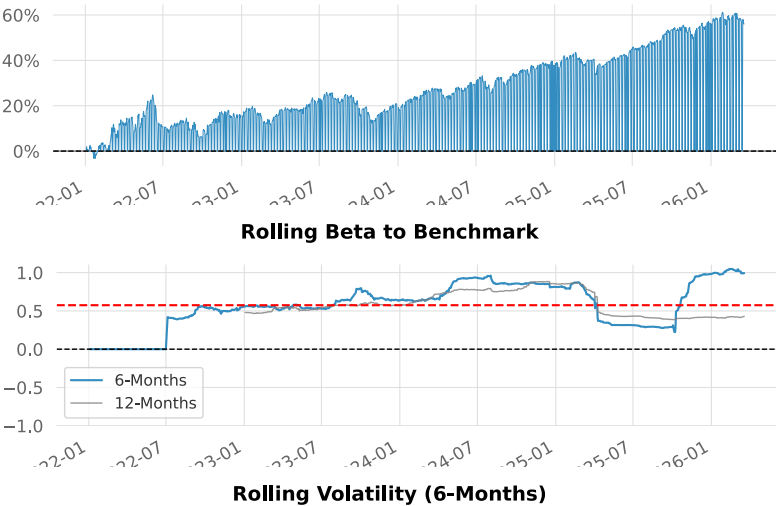
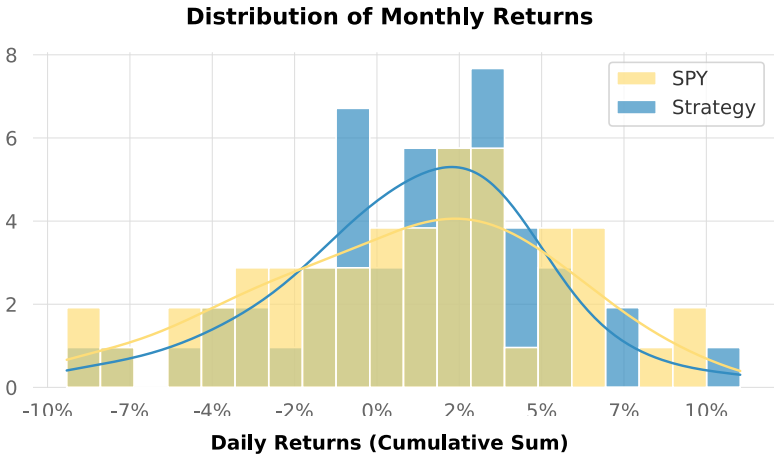
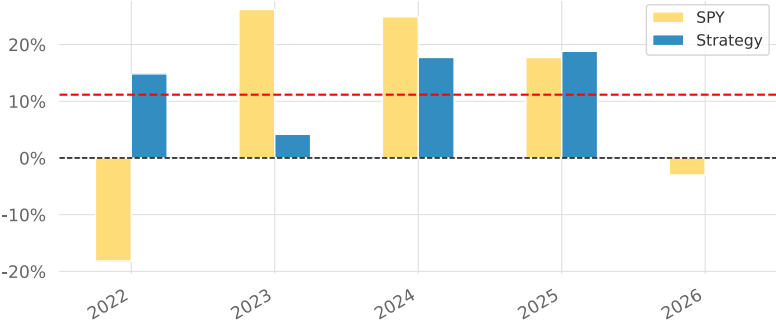
Benchmark: SPY • Periods/Year: 252 • RF: 0.0% | Generated by [QuantStats](#) (v. 0.0.81)



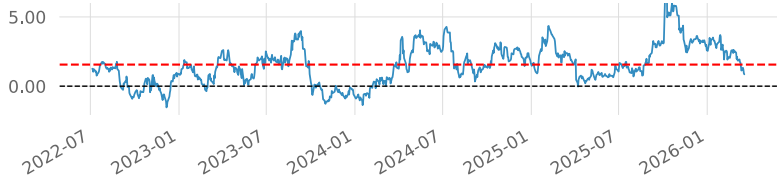
Key Performance Metrics

Metric	SPY	Strategy
Risk-Free Rate	0.0%	0.0%
Time in Market	100.0%	100.0%
Cumulative Return	47.22%	67.72%
CAGR %	9.67%	13.13%
Sharpe	0.61	0.94
Prob. Sharpe Ratio	89.46%	97.02%
Smart Sharpe	0.6	0.93
Sortino	0.88	1.31
Smart Sortino	0.87	1.3
Sortino/√2	0.62	0.93
Smart Sortino/√2	0.62	0.92
Omega	1.12	1.18
Max Drawdown	-24.5%	-17.27%
Max DD Date	2022-10-12	2022-09-26
Max DD Period Start	2022-01-04	2022-06-08
Max DD Period End	2023-12-12	2024-03-06
Longest DD Days	708	638
Volatility (ann.)	17.76%	14.26%
R ²	0.43	0.43
Information Ratio	0.01	0.01
Calmar	0.39	0.76
Skew	0.36	-0.56
Kurtosis	9.26	4.18
Ulcer Performance Index	5.15	11.03
Risk-Adjusted Return	9.67%	13.13%
Risk-Return Ratio	0.04	0.06
Avg. Return	0.04%	0.05%
Avg. Win	0.81%	0.68%

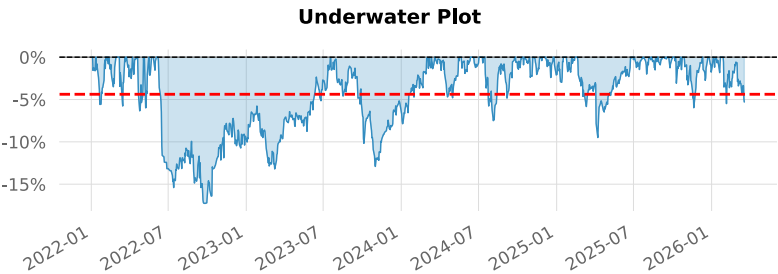
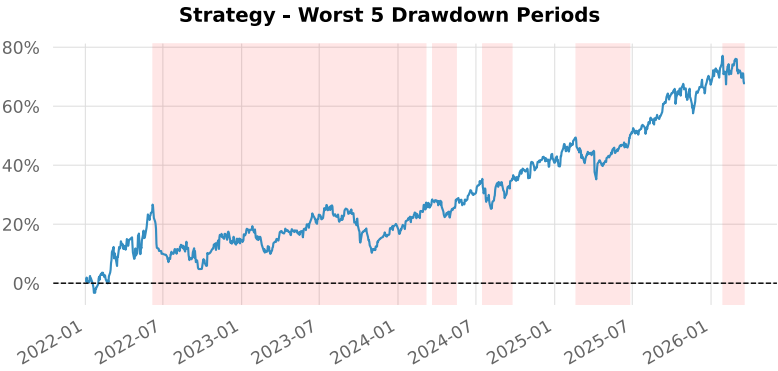
EOY Returns vs Benchmark



Metric	SPY	Strategy
Avg. Loss	-0.86%	-0.74%
Win/Loss Ratio	0.95	0.91
Profit Ratio	0.82	0.74
Expected Daily	0.04%	0.05%
Expected Monthly	0.76%	1.02%
Expected Yearly	8.04%	10.9%
Kelly Criterion	4.81%	6.87%
Risk of Ruin	0.0%	0.0%
Daily Value-at-Risk	-1.8%	-1.42%
Expected Shortfall (cVaR)	-2.74%	-2.11%
Max Consecutive Wins	10	10
Max Consecutive Losses	6	10
Gain/Pain Ratio	0.12	0.18
Gain/Pain (1M)	0.61	1.06
Payoff Ratio	0.95	0.91
Profit Factor	1.12	1.18
Common Sense Ratio	1.13	1.12
CPC Index	0.57	0.6
Tail Ratio	1.01	0.95
Outlier Win Ratio	3.4	3.43
Outlier Loss Ratio	3.84	3.61
MTD	-3.58%	-4.74%
3M	-1.93%	1.76%
6M	0.45%	3.51%
YTD	-3.0%	0.25%
1Y	19.28%	16.82%
3Y (ann.)	18.94%	12.87%
5Y (ann.)	9.67%	13.13%
10Y (ann.)	9.67%	13.13%
All-time (ann.)	9.67%	13.13%
Best Day	10.5%	4.08%
Worst Day	-5.85%	-5.99%
Best Month	9.21%	11.03%
Worst Month	-9.24%	-9.42%



Metric	SPY	Strategy
Best Year	26.18%	18.81%
Worst Year	-18.18%	0.25%
Avg. Drawdown	-2.07%	-2.51%
Avg. Drawdown Days	24	26
Recovery Factor	1.85	3.24
Ulcer Index	0.09	0.06
Serenity Index	0.28	0.61
Avg. Up Month	4.03%	3.51%
Avg. Down Month	-4.36%	-3.26%
Win Days	53.66%	55.57%
Win Month	62.75%	62.75%
Win Quarter	64.71%	76.47%
Win Year	60.0%	100.0%
Beta	-	0.53
Alpha	-	0.08
Correlation	-	65.63%
Treynor Ratio	-	128.52%



2022	-0.33	3.96	7.68	-1.98	11.03	-9.42	1.47	-2.63	-3.52	7.76	3.85	-2.13
2023	2.88	-6.23	5.88	0.87	-0.29	4.45	2.56	-1.08	-4.21	-7.56	5.09	2.82
2024	0.89	3.46	2.65	-4.41	3.56	2.61	-0.18	3.33	1.18	-0.23	4.58	-0.66
2025	3.34	0.20	-0.94	-2.13	2.87	4.03	0.62	2.64	5.03	1.04	-0.51	1.42
2026	2.12	3.05	-4.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC



EOY Returns vs Benchmark

Year	SPY	Strategy	Multiplier	Won
2022	-18.18%	14.83%	-0.82	+
2023	26.18%	4.17%	0.16	-
2024	24.89%	17.72%	0.71	-
2025	17.72%	18.81%	1.06	+
2026	-3.00%	0.25%	-0.08	+

Worst 10 Drawdowns

Started	Recovered	Drawdown	Days
2022-06-08	2024-03-06	-17.27%	638
2025-02-20	2025-06-25	-9.49%	126
2024-07-17	2024-09-23	-7.50%	69
2022-04-21	2022-05-03	-6.34%	13
2022-05-09	2022-05-16	-5.99%	8

Started	Recovered	Drawdown	Days
2025-10-29	2025-12-09	-5.97%	42
2022-03-09	2022-03-23	-5.77%	15
2022-01-13	2022-02-03	-5.58%	22
2026-01-29	2026-03-19	-5.48%	50
2024-03-22	2024-05-16	-4.80%	56